entangledECAR

Entangled cells in Reversible Elementary Cellular Automata

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1 What the program does

`entangledECAR` is a MATLAB script related to ECARdoku that generates visualisations of a reversible elementary cellular automaton (Rule 37R) in a *not fully determined regime*.

The program restricts the solution space one clue at a time. Instead of forcing a *unique* solution (as in classical Sudoku-style puzzles), it deliberately stops when exactly a small user-chosen number of solutions (typically 4–8) remain. At that point:

- The **determined** cells (always black or always white) have the same value in **all** retained solutions.
- The **indeterminate** cells take different values across the remaining solutions.
- Among the indeterminate cells, the program finds all **anticorrelated pairs** (pairs of cells whose values are always opposite in every remaining solution).
- Correlated indeterminate cells and fully determined cells are left uncoloured (grey, white, or black).
- Each anticorrelated pair is coloured with a distinct bright colour.

The result is a visual representation of apparent entanglement arising purely from global constraints in a perfectly classical, local, reversible cellular automaton.

2 How it works

1. **Precomputation (once)**: All 2^n possible initial rows are evolved for m generations using reversible Rule 37R (periodic boundaries + XOR carry term). The boards are stored as a 3-D `uint8` array.
2. **Interactive clue loop**: The user chooses a target number of solutions $N \geq 2$. Clues are added either manually or randomly until exactly N boards remain. The clues are saved in the text file `clues.txt`.
3. **Pair detection**: All pairs of indeterminate cells are examined. Only **anticorrelated** pairs are retained.
4. **Colouring**: Up to 8 anticorrelated pairs are selected and assigned distinct bright colours. All other cells stay white (0), grey (indeterminate), or black (1).
5. **Output**: A clean high-resolution PNG (100 pixels per cell) and the on-screen MATLAB figure are generated.

Important note on random mode: When using random clues, the program can occasionally enter a very long loop (or appear stuck) while searching for a set of clues that reduces the number of possible boards to exactly the requested target. This risk is higher for large target numbers (e.g. 100+). It usually does not happen when the chosen number of solutions is small (e.g. 4–12).

3 Usage

- Open `entangledECAR.m` in MATLAB.
- Run the script.
- Answer the on-screen prompts (manual or random clues, target number of solutions).
- The script will add clues, compute the remaining solutions, colour the anticorrelated pairs, and save the clean PNG.
- Repeat the target prompt as many times as you like (press Enter to exit).

4 Metaphysical musings

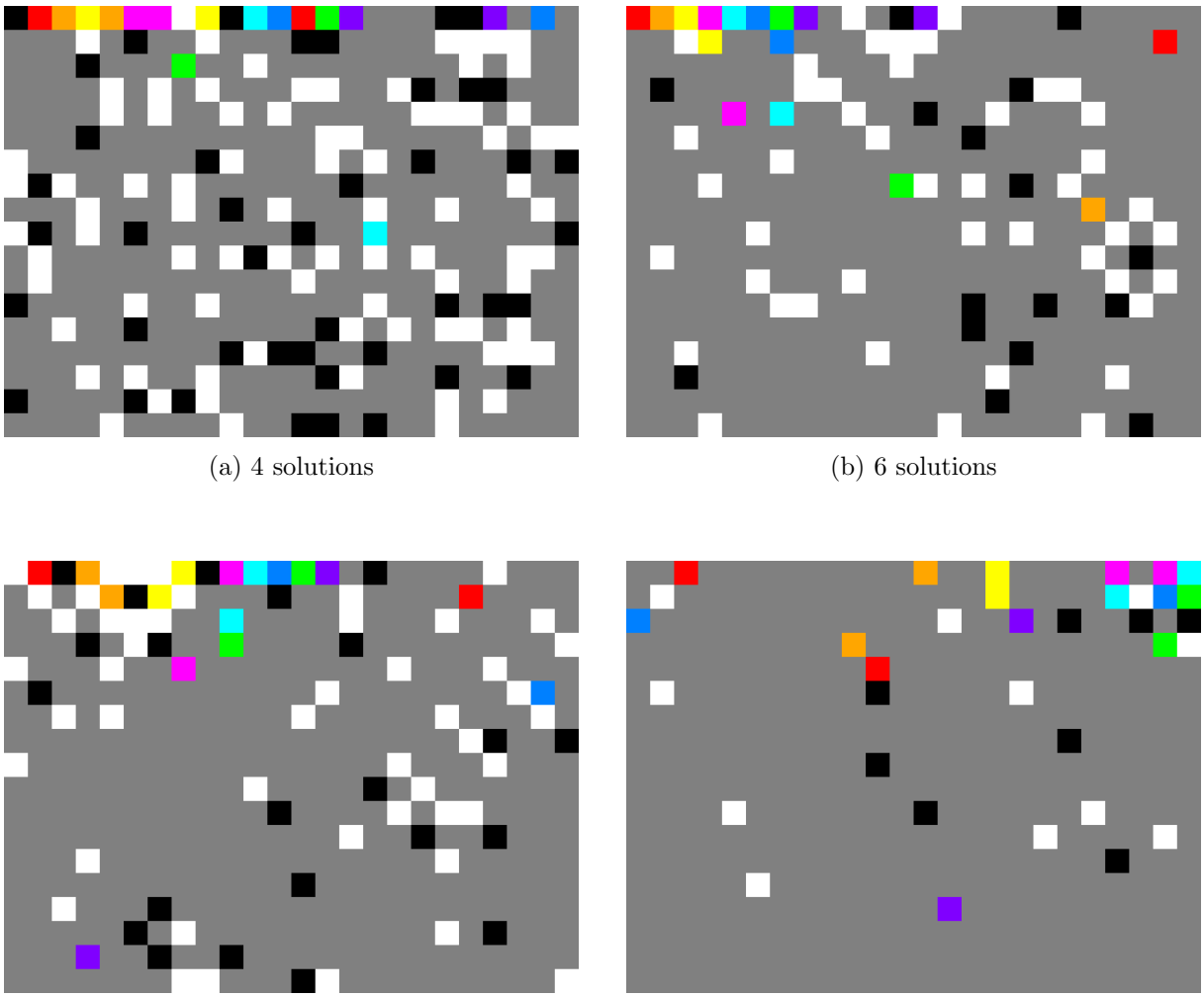
This program is an illustration of entanglement and of the deeper concept that the universe's choices (the clues) constrain what is possible. Choosing a value for one cell may instantly remove one of the possibilities for another cell — even one that is far away on the board — without any local interaction or signal passing between them. Or, two other cells may remain undetermined but become correlated or anticorrelated (entangled). Remarkably, a cell can be entangled not only with another cell in its present, but also with a cell in its future. This is clearly visible in the first row of the plots below (see Figure 1), where some cells are anticorrelated with cells several generations below them.

A natural extension of the model is a growing grid in which two new indeterminate cells are added to the left and right edges at every time step. In this version the universe is literally expanding: at each new row fresh possibilities are created at the horizon while existing possibilities are continuously pruned by the clues. A clue given late in the simulation, even if given on one of the newly added edge cells, can retroactively force an anticorrelation with a cell that existed many generations earlier. It can also create a perfect anticorrelation between an early cell and a cell that is added much later. This makes the two distinct notions of time especially vivid: the internal *simulation time* (the vertical evolution of the grid) and the external *time of the simulating program* (the chronological order in which clues are successively given). The simulation can therefore entangle a cell with its own future.

The apparent non-locality is purely the result of global constraints on the set of allowed realities. The underlying physics remains completely classical and local; only our knowledge of the system becomes non-local once the clues are applied. In this sense, entangledECAR shows how entanglement-like correlations can emerge from nothing more than the progressive restriction (and creation) of possibility space.

5 Example visualizations

The following figures show typical outputs for 24×24 boards (displayed cropped to 24×18 for readability) with 4, 6, 8 and 12 remaining solutions. Only anticorrelated pairs are highlighted with bright colours.



(a) 4 solutions

(b) 6 solutions

(c) 8 solutions

(d) 12 solutions

Figure 1: 24×24 boards (cropped to 24×18) showing only anticorrelated pairs in bright colours.